

CLIENT REVIEW DOCUMENT

Rahma AI Guardian App Architecture and Implementation Plan

A practical Phase 1 and Phase 2 delivery plan covering system architecture, data flow, UX flow, server architecture, open technical options, recommended decisions, and launch readiness.

Prepared: 22 May 2026

Phase 1: Elderly and special-care pilot

Phase 2: Children, geofencing, pharmacy, languages

1. Executive Summary

Rahma is planned as a one-touch emergency and care-assistance platform for vulnerable users. Phase 1 should prove the emergency-care foundation for elderly and special-care users. Phase 2 should expand the platform into children, geofencing, pharmacy communication, broader language support, and selected wearable integration.

Recommended build model

Native iOS and Android apps, reusable backend services, receiver dashboard, and client-approved UAE-hosted infrastructure.

Recommended hosting

AWS UAE Region with a UAE-hosted foundation-model layer. Custom on-premises hosting should be treated as a separate proposal.

Recommended AI posture

AI supports language and intent context. The one-touch activation itself starts the safety workflow.

We will meet any compliance, audit, data handling, or security requirement specifically confirmed by the client or authority stakeholders. Until those requirements are formally shared, the architecture will keep controls configurable and evidence-ready.

2. Confirmed Scope

Phase 1: Elderly and Special-Care Emergency Pilot

- One-touch emergency activation with press-and-hold protection.
- Voice and visual yes/no interaction running in parallel.
- Arabic and English support.
- Emergency event creation with timestamp, location, profile summary, and status.
- Caregiver/family notification through push and SMS fallback.
- Receiver dashboard for authority/support/family users.
- Hospital finder using an approved facility directory.
- Medication reminders for the user.
- Offline and weak-network fallback using local data, last known location, SMS, and later sync.
- Audit trail for emergency events, receiver actions, notification status, and support handling.

Phase 2: Children, Geofencing, Pharmacy, Scale, and Languages

- Child profile module with parent/guardian access.
 - School or authorized receiver workflows.
 - Safe-zone/geofencing setup and alerts.
 - Family member messaging and escalation.
 - Pharmacy communication for prescription/dosage supply workflows.
 - Medication reminder expansion with missed-dose escalation.
 - Multilingual expansion beyond Arabic and English.
 - Wearable integration as a controlled workstream, depending on selected devices and APIs.
-

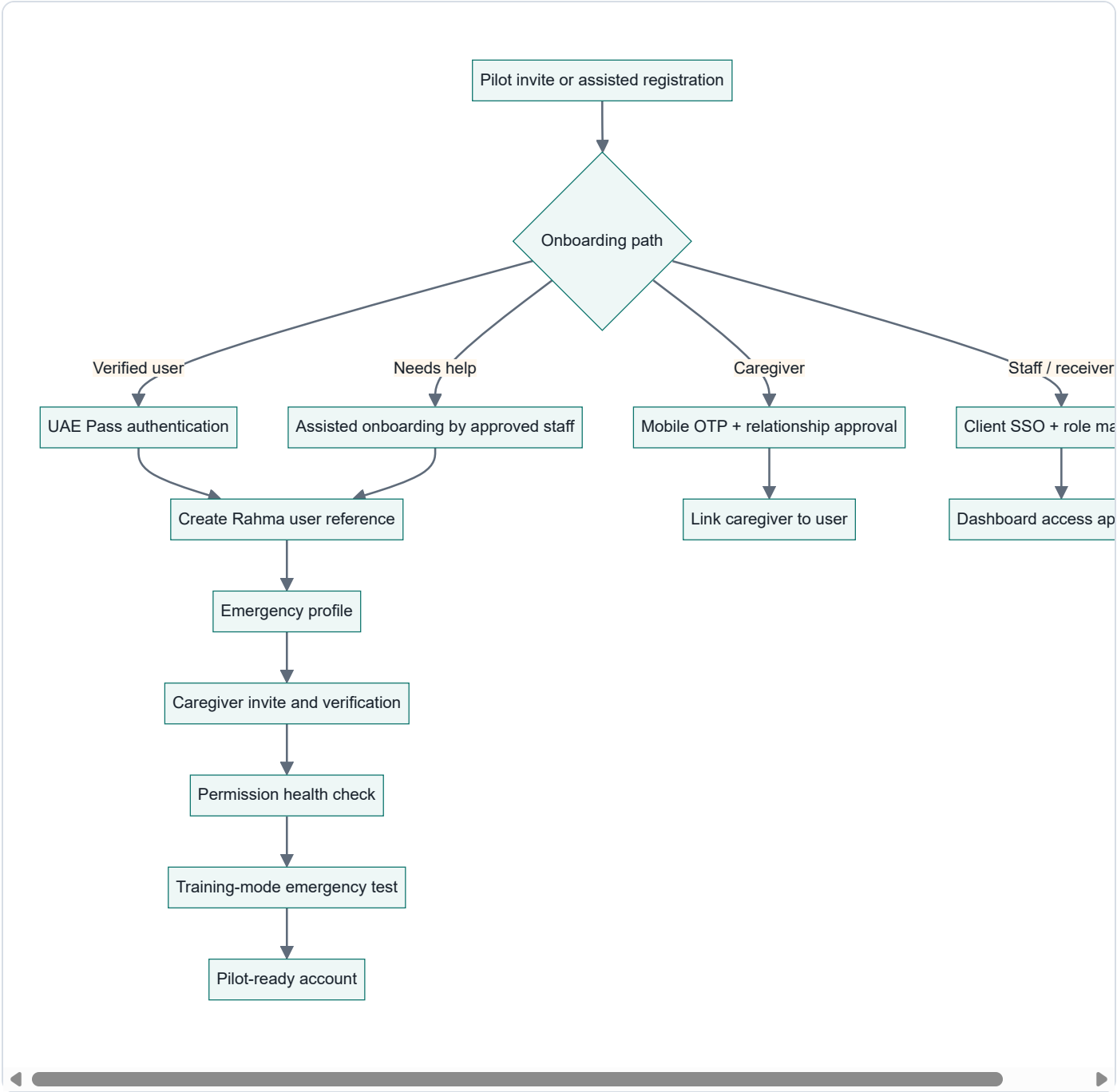
3. Recommended User Onboarding Flow

Rahma onboarding should support verified identity while staying practical for elderly and special-care users. Our recommendation is UAE Pass as the primary route where approved, assisted onboarding as the elderly-friendly fallback, mobile OTP plus relationship approval for caregivers, and client SSO for staff/dashboard users.

Option	Best fit	Recommendation
UAE Pass	Verified identity for eligible Phase 1 users.	Recommended primary
Assisted onboarding	Elderly or special-care users who need staff/caregiver help.	Recommended fallback
Mobile OTP	Caregivers and family members.	Use with relationship a
Client/staff SSO	Admins, support teams, and receiver-dashboard users.	Recommended for staf
Social login	Low-risk convenience flows.	Not recommended for I

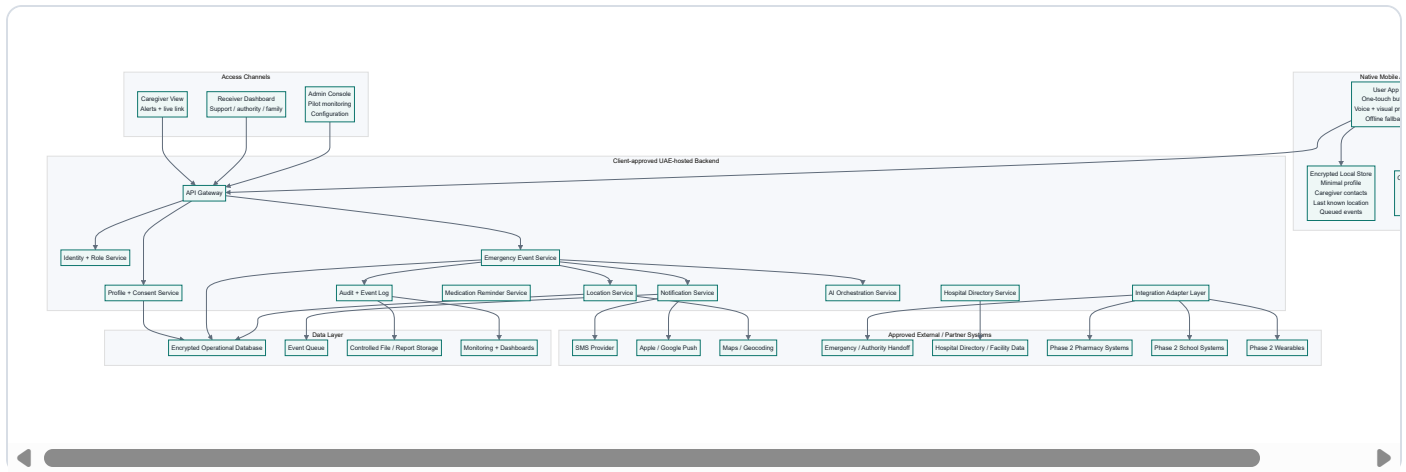
Recommended Phase 1 Onboarding Steps

1. Pilot invite or staff-assisted registration starts the process.
2. User chooses UAE Pass, assisted onboarding, or approved manual pilot setup.
3. Identity is verified and a Rahma user reference is created.
4. Minimal emergency profile is completed: language, medical flags, allergies, caregiver contacts, and medication reminders.
5. Caregivers receive invite and verify by OTP plus relationship approval.
6. App performs permission health check for location, notifications, microphone, and SMS fallback.
7. User completes a guided emergency test in training mode.
8. Account becomes pilot-ready only after profile, caregiver, permissions, and test status are complete.



4. Recommended High-Level Architecture

Rahma should be implemented as a secure mobile-first platform with five layers: native mobile apps, receiver interfaces, backend platform services, AI/voice services, and integration adapters.



5. How the App Works: Activation to Emergency Handoff

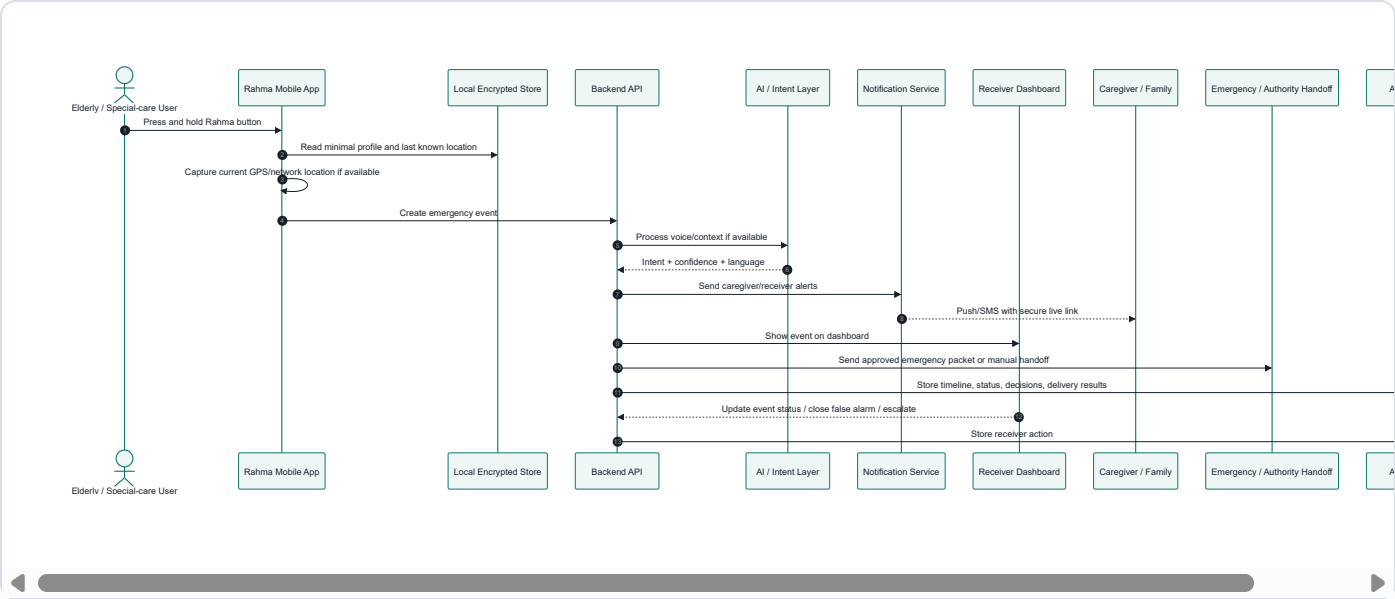
Rahma should be explainable to the client as a simple emergency workflow. The user presses and holds the main Rahma button for 2 seconds, the app starts the emergency workflow, captures location and minimal profile context, alerts caregivers/receivers, and then dispatches or hands off through the approved route.

1. **Activate Rahma:** User presses and holds the main button for 2 seconds.
2. **Confirm context:** App asks one simple voice/visual prompt such as Are you hurt? or Do you need help?
3. **Capture location and profile:** App captures GPS/network location or falls back to last-known location with timestamp and confidence.
4. **Create emergency event:** Backend creates a timestamped emergency event with activation source, user reference, location quality, and workflow status.
5. **Alert caregivers and receivers:** Push/SMS alerts and dashboard events start immediately, with a normal-connectivity target of under 10 seconds.
6. **Dispatch or handoff:** If a direct approved authority API exists, the emergency packet is sent through that route. If not, Phase 1 uses receiver dashboard, call-center/manual handoff, or simulator.
7. **Track and close:** Receivers acknowledge, escalate, mark false alarm, or close the event. Every action is audited.

Recommended authority dispatch position: use the client-confirmed authority or emergency API route wherever it is available. Build the emergency packet, receiver dashboard, and audit trail as the shared foundation for every handoff route, while keeping approved dashboard, call-center, manual handoff, or simulator options as operational fallback paths during pilot validation.

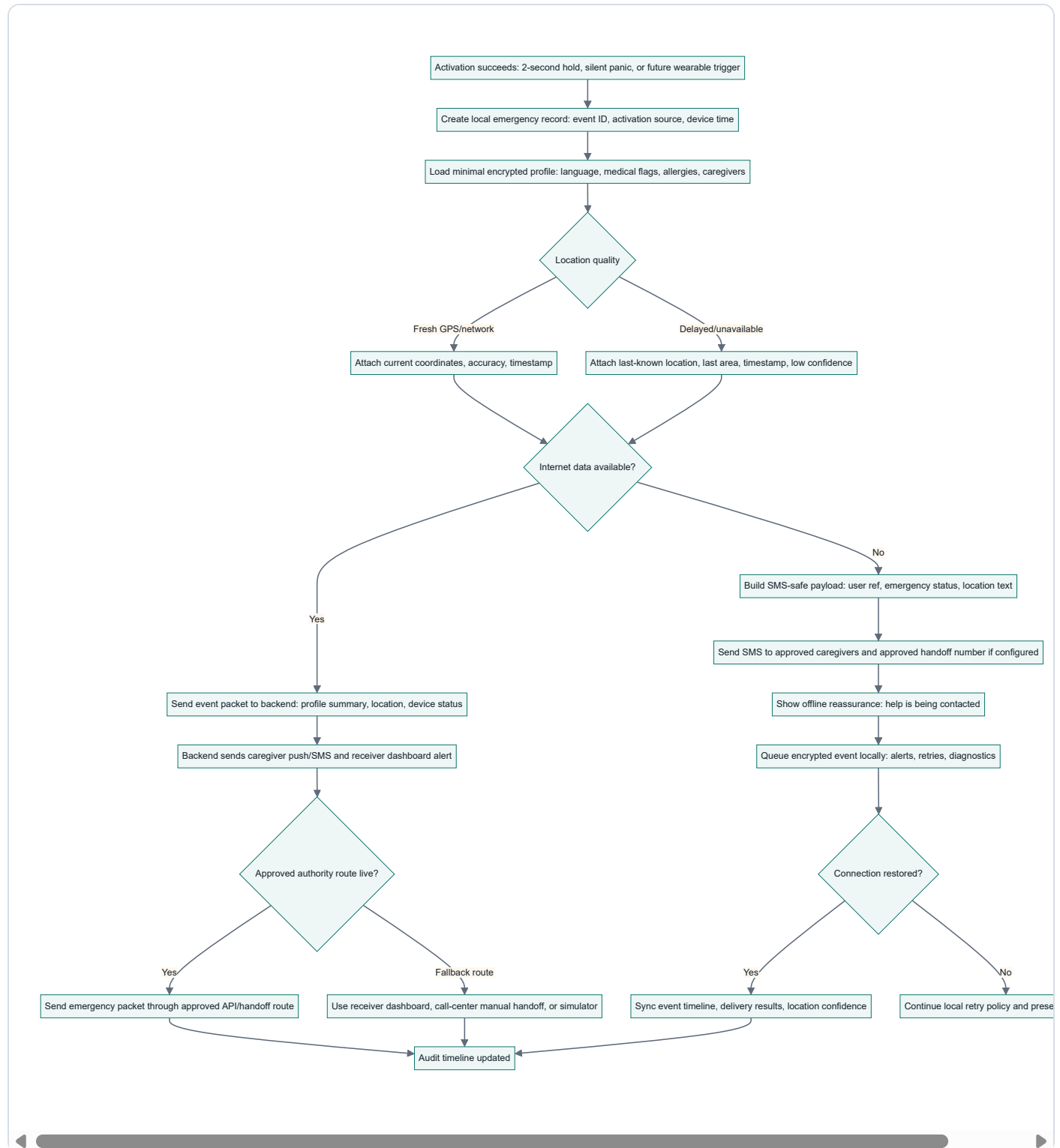
6. Phase 1 Emergency Flow

The Phase 1 flow must be usable by an elderly or special-care user during distress. No typing or menu navigation should be required.



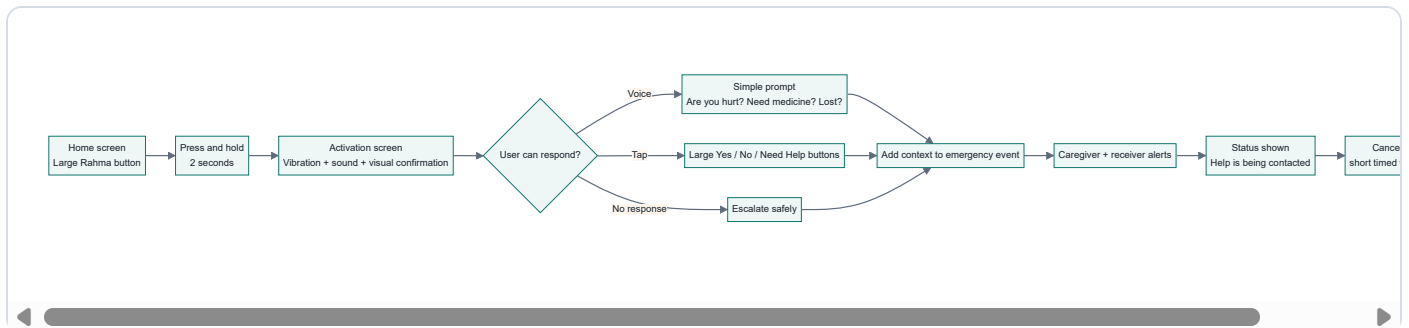
7. Help Must Start Even When the Internet Is Imperfect

Offline mode should focus on the emergency-critical path and show exactly what happens when internet, GPS, push notifications, or authority APIs are unavailable.



Full offline medical triage is not recommended for Phase 1. The offline model should support simple commands, language handling, and yes/no prompts while the button activation starts the emergency workflow.

8. User Experience Flow

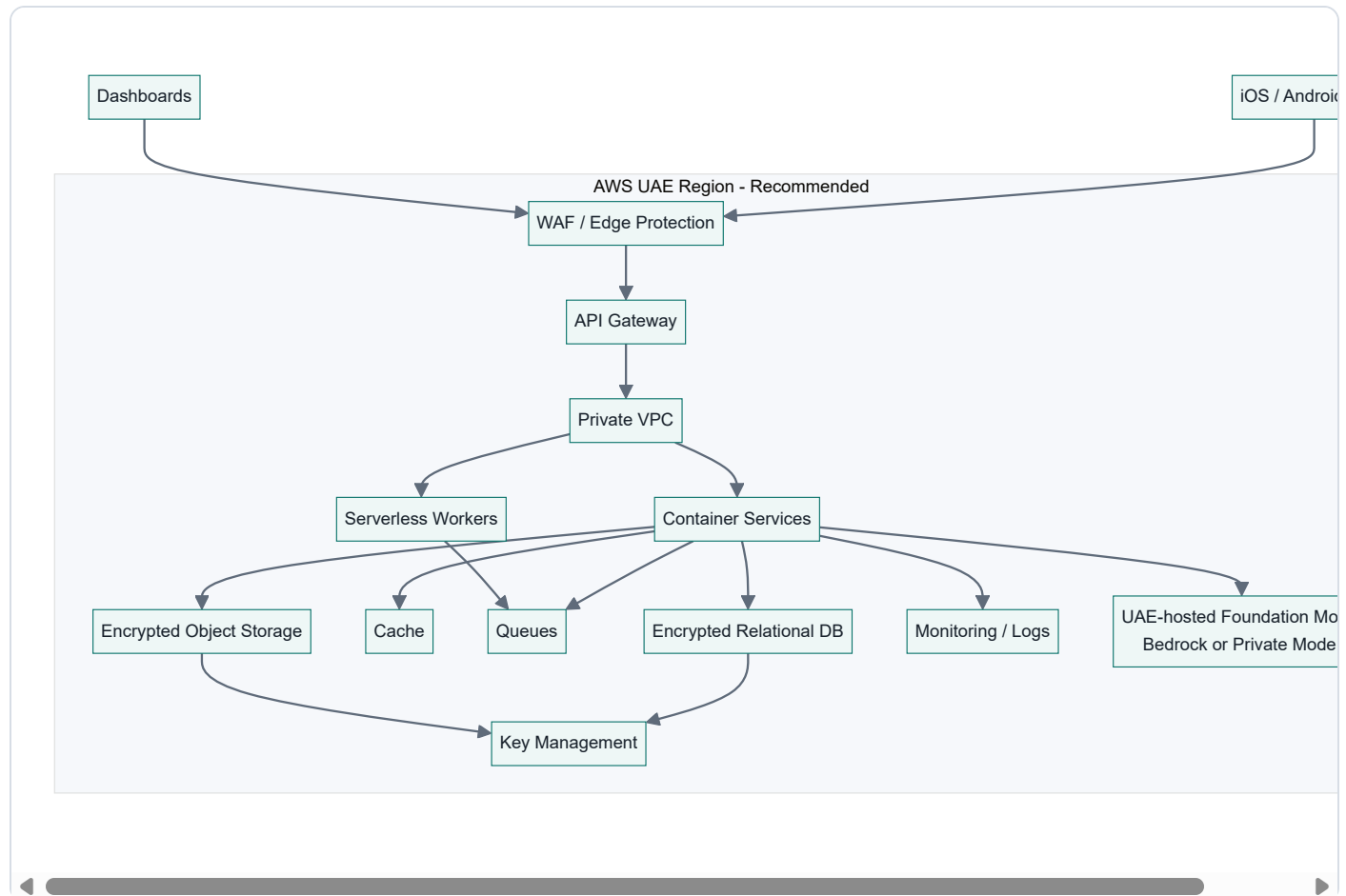


- One primary action on the home screen.
- Voice and visual buttons available at the same time.
- Haptic and audio feedback for reassurance.
- Large text, high contrast, simple Arabic/English language.
- Clear status after activation.

9. Recommended Server Architecture

Preferred Option: AWS UAE Region with UAE-Hosted Foundation Model

This is the recommended option from our side for the Phase 1 pilot. It is faster for Phase 1, easier to operate, and better suited for scaling into Phase 2 than building a custom GPU environment from day one. A UAE-hosted foundation model can support higher-quality Arabic/English understanding while keeping sensitive patient data inside the approved hosting boundary.



Alternative: Custom On-Premises / Dedicated GPU Server

Use this only if the client requires a fully dedicated environment. It provides maximum infrastructure control but increases cost, procurement time, MLOps effort, security maintenance, and delivery risk.

Alternative: Hybrid

Use managed cloud for app services and a private AI server for selected sensitive model workloads. This can balance speed and control, but it adds integration complexity.

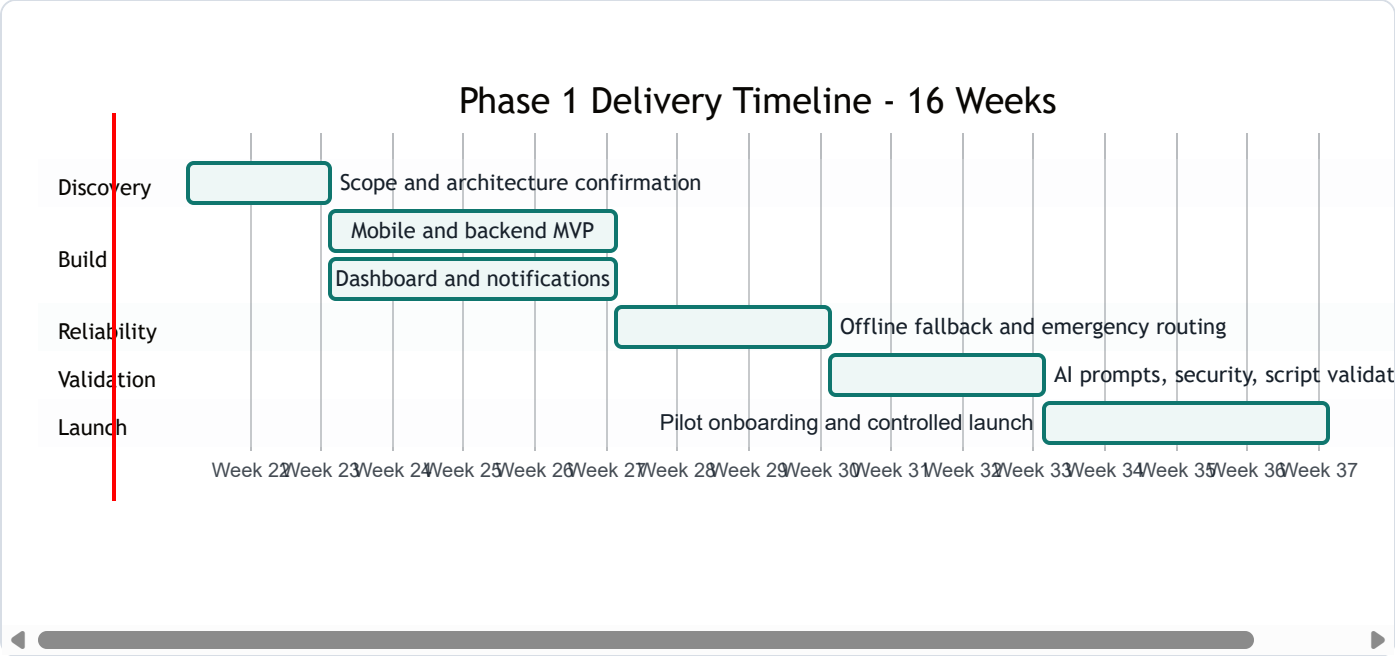
10. Open Decisions and Recommended Options

Topic	Options	Re
Mobile technology	Native iOS/Android, Flutter, React Native	Na
Hosting	AWS UAE Region + UAE-hosted foundation model, custom on-prem, hybrid	AW
AI model usage	UAE-hosted foundation model, UAE-hosted private model, self-hosted model	Usi
Offline voice	Lightweight on-device model, rules-only prompts, full offline AI triage	Lig
Emergency integration	Direct approved API, gateway/call-center, simulator/manual pilot	Usi
Onboarding	UAE Pass, mobile OTP, client SSO, assisted onboarding	UA
Medication reminders	App reminders, caregiver escalation, pharmacy-connected reminders	Phi
Wearables	Defer, Phase 2 proof, full launch support	Phi

11. Phase 1 Implementation Plan

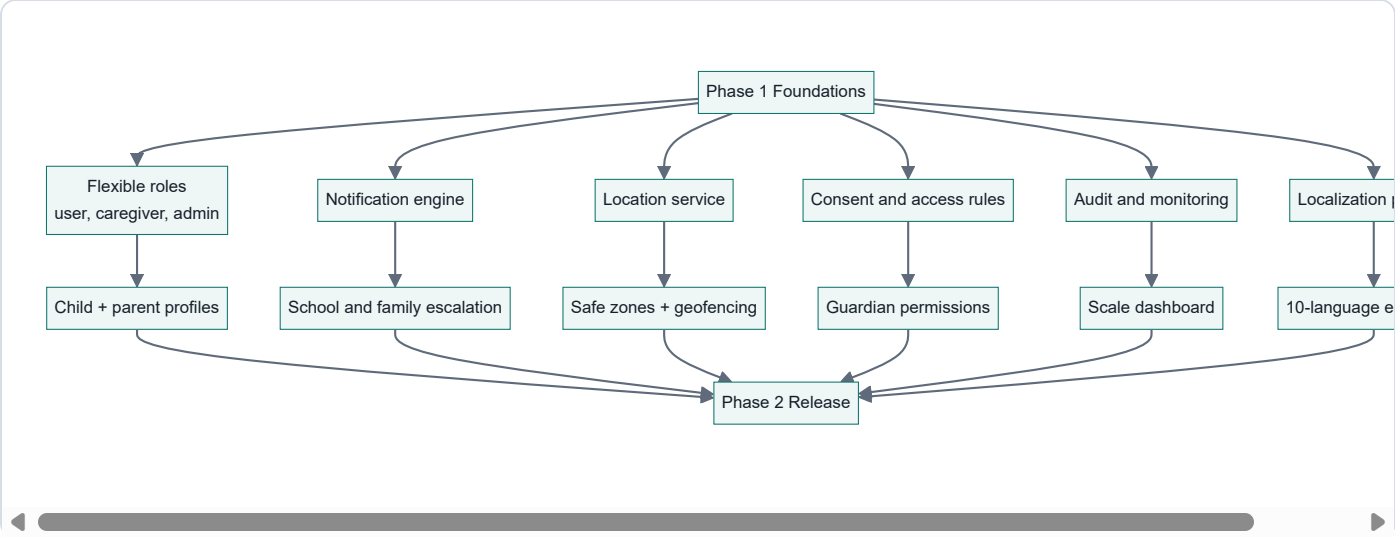
Recommended duration: 16 weeks, assuming client approvals, hosting access, and integration inputs are available on time.

Period	Focus	Key outputs
Weeks 1-2	Discovery and scope lock	Final scope, hosting decision, onboarding decision, emer
Weeks 3-6	Core app and backend MVP	Native app MVP, profile service, emergency service, care
Weeks 7-9	Offline and emergency reliability	SMS fallback, local queue, last-known location, receiver I
Weeks 10-12	AI, security, and script validation	Arabic/English prompts, intent confidence policy, approve
Weeks 13-16	Pilot readiness and launch	Production deployment, pilot onboarding, dashboards, inc



12. Phase 2 Implementation Plan

Phase 2 should start after Phase 1 validates activation success, notification delivery, location quality, false alarm handling, support workload, dashboard usability, hosting readiness, and identity/access foundations.



13. Data Flow and Storage

Data category	Example	Storage approach
Identity data	User reference, caregiver reference	Encrypted database
Emergency profile	Medical flags, allergies, language	Encrypted database and minimal encr
Location data	Current/last known GPS, timestamp, accuracy	Event-linked storage
Emergency event	Activation source, status, timeline	Event database + audit log
Voice/transcripts	Intent and confidence	Store only derived metadata by default

14. Risks and Mitigation

Risk	Impact	Mitigation
Emergency API access is delayed	Pilot launch blocked if direct integration is mandatory	Build simula
AI misunderstands distressed speech	Wrong context or delay	Button activa
Weak GPS indoors	Incomplete location	Use GPS, no
Internet unavailable	Push/API path fails	Use local ev
Caregiver access misconfigured	Privacy and trust issue	Verify caregi
Custom hosting required	Higher cost and timeline	Prepare sep

15. Immediate Client Decisions

1. Hosting preference: AWS UAE Region with a UAE-hosted foundation-model layer, dedicated on-premises infrastructure, or hybrid.
2. Onboarding preference: UAE Pass, mobile OTP, client SSO, or assisted/manual onboarding.
3. Patient information data access method linked to UAE Pass: approved API, secure client integration, controlled data feed, or manual pilot import.
4. Emergency handoff model for Phase 1.
5. SMS payload approval for fallback alerts.
6. Receiver roles for Phase 1 alerts.
7. Approved hospital directory source.
8. Medication reminder scope for Phase 1.
9. Voice/audio retention policy.
10. Phase 2 priority order.
11. Any specific audit, security, data residency, or documentation requirements to be met.

16. Final Recommendation

Rahma should proceed with a disciplined Phase 1 pilot focused on emergency reliability, accessibility, location capture, caregiver/receiver alerting, medication reminders, auditability, and operational readiness.

The most favorable technical path is native iOS and Android apps, AWS UAE Region with a UAE-hosted foundation-model layer, constrained on-device offline logic, SMS fallback, reusable backend services, and a receiver dashboard. This gives the client a realistic foundation while keeping Phase 2 expansion possible without rebuilding the platform.